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QUESTION 1:

***DRAW A DIAGRAM OF THE OSI 7-LAYER
MODEL AND LABEL EACH LAYER WITH
ITS CORRECT NAME AND A REAL-WORLD
EXAMPLE OF A DEVICE OR PROTOCOL
THAT OPERATES AT THAT LAYER (E.G.,
SWITCH AT DATA LINK, HTTP AT
APPLICATION).***

ANSWER:

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*/ 7. APPLICATION LAYER / HTTP,
HTTPS, FTP*

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*/ 6. PRESENTATION LAYER / SSL/TLS
ENCRYPTION*

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/ 5. SESSION LAYER / NETBIOS, RPC

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/ 4. TRANSPORT LAYER / TCP, UDP

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/ 3. NETWORK LAYER / ROUTER, IP

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*/ 2. DATA LINK LAYER / SWITCH, MAC
ADDRESS*

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/ 1. PHYSICAL LAYER / CABLE, HUB

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QUESTION 2:

*MATCH THE FOLLOWING NETWORKING
SCENARIOS TO THE CORRECT OSI
LAYER:*

A) ENCRYPTING DATA BEFORE SENDING

*B) ROUTING PACKETS ACROSS
NETWORKS*

*C) CONVERTING DATA TO ELECTRICAL
SIGNALS*

*D) OPENING A WEB PAGE IN YOUR
BROWSER*

*E) DETECTING AND CORRECTING
ERRORS IN FRAMES*

*HINT: WRITE YOUR ANSWERS AS
'SCENARIO: LAYER NAME (LAYER
NUMBER)'.*

ANSWER:

*A) ENCRYPTING DATA BEFORE
SENDING: PRESENTATION LAYER
(LAYER 6)*

*B) ROUTING PACKETS ACROSS
NETWORKS: NETWORK LAYER (LAYER
3)*

*C) CONVERTING DATA TO ELECTRICAL
SIGNALS: PHYSICAL LAYER (LAYER 1)*

*D) OPENING A WEB PAGE IN YOUR
BROWSER: APPLICATION LAYER
(LAYER 7)*

*E) DETECTING AND CORRECTING
ERRORS IN FRAMES: DATA LINK LAYER
(LAYER 2)*

QUESTION 3:

*PICK ANY APP YOU USE DAILY (LIKE
WHATSAPP, INSTAGRAM, OR ZOMATO)
AND DESCRIBE HOW AT LEAST THREE
DIFFERENT OSI LAYERS ARE INVOLVED
WHEN YOU SEND OR RECEIVE DATA IN
THAT APP.*

ANSWER (WHATSAPP EXAMPLE):

*WHEN I SEND A MESSAGE ON
WHATSAPP:*

*APPLICATION LAYER (LAYER 7) –
WHATSAPP CREATES AND DISPLAYS
THE MESSAGE.*

*TRANSPORT LAYER (LAYER 4) – TCP
ENSURES THE MESSAGE IS DELIVERED
CORRECTLY.*

NETWORK LAYER (LAYER 3) – IP ADDRESSES HELP ROUTE THE MESSAGE THROUGH THE INTERNET.

DATA LINK LAYER (LAYER 2) – WI-FI OR MOBILE NETWORK TRANSFERS THE DATA.

PHYSICAL LAYER (LAYER 1) – DATA TRAVELS AS RADIO WAVES OR ELECTRICAL SIGNALS.

QUESTION 4:

GIVEN THIS SCENARIO: YOU ARE UNABLE TO ACCESS A WEBSITE, BUT YOU CAN SUCCESSFULLY PING ITS IP ADDRESS. IDENTIFY WHICH OSI LAYER(S) MIGHT BE CAUSING THE PROBLEM AND EXPLAIN YOUR REASONING.

ANSWER:

THE PROBLEM IS LIKELY IN THE APPLICATION LAYER (LAYER 7),

*PRESENTATION LAYER (LAYER 6), OR
SESSION LAYER (LAYER 5).*

REASON:

*PING WORKS, WHICH MEANS THE
NETWORK LAYER (LAYER 3) IS
FUNCTIONING CORRECTLY.*

*THE DEVICE CAN REACH THE SERVER'S
IP ADDRESS.*

THE ISSUE MAY BE DUE TO:

DNS PROBLEMS

HTTP/HTTPS SERVICE NOT WORKING
SSL/TLS CERTIFICATE ISSUES
BROWSER CONFIGURATION PROBLEMS

*CONCLUSION: SINCE THE NETWORK
CONNECTION IS WORKING, THE PROBLEM
IS MOST LIKELY IN LAYER 5, LAYER 6,
OR LAYER 7.*